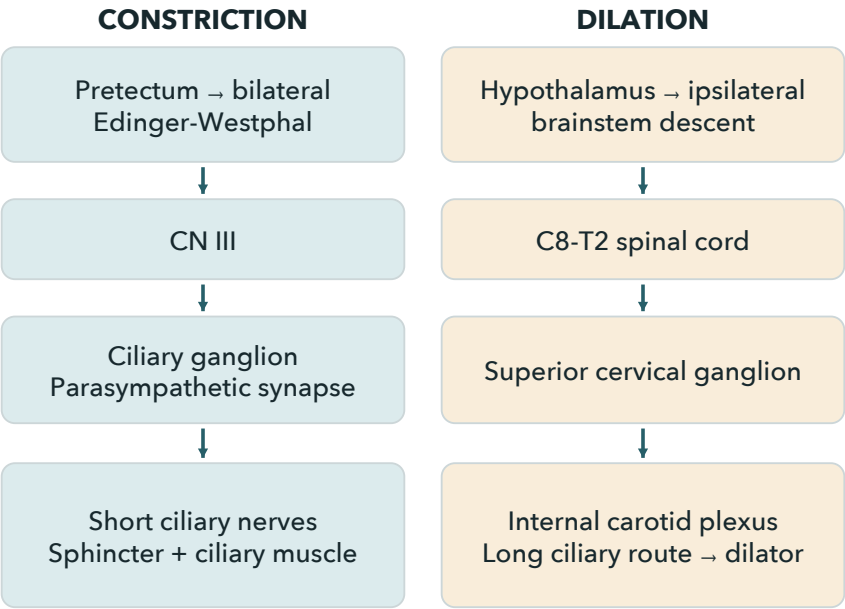


Autonomic pupil circuits

Trace constriction and dilation separately, then compare both eyes.

Two motor routes, different synapses

A | Parasympathetic and sympathetic routes



Muscarinic

Alpha-1 dilator

Sympathetic descent is ipsilateral. Sensory and sympathetic fibres traverse the ciliary ganglion without synapsing.

Targets and receptors

Route	Receptor	Effect
Parasympathetic	Muscarinic	Sphincter and ciliary muscle contract
Sympathetic	Alpha-1 at dilator	Radial dilator contracts
Ciliary ganglion	Nicotinic	Parasympathetic ganglion cells activated

Fast check

After the superior cervical ganglion, ocular fibres follow the internal carotid route; many facial sweating fibres follow the external carotid route. Sweating loss therefore varies with lesion level.

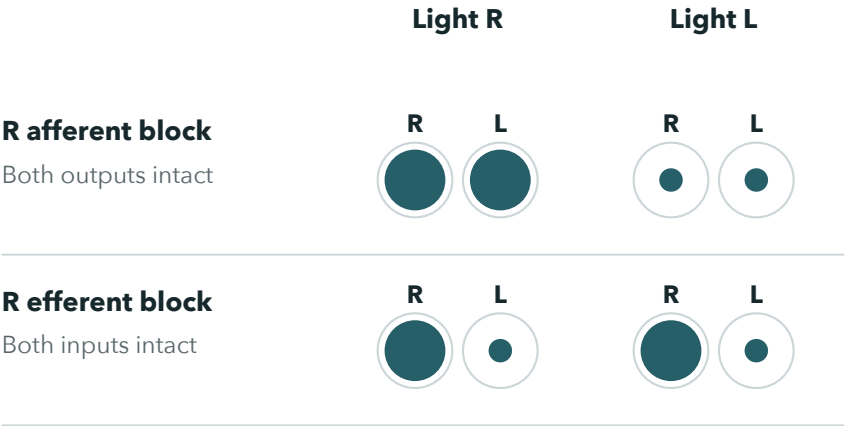
Figure reference

A: pupil reflex and Horner pathways. Original schematic.

Stimulus versus effector failure

B | Read both pupils under both stimuli

Responses to each light stimulus



Large = no constriction | small = constriction

Normal left constriction with right-eye illumination demonstrates usable right afferent input in the efferent example.

Add the near response

Pattern	Mechanism clue
Poor light, better near	Different premotor drives can reach shared final effectors
Slow near constriction and redilation	Postganglionic denervation/reinnervation can produce a tonic response
Stronger response to dilute agonist	Denervation hypersensitivity supports the mechanism in context

Fast check

Compare anisocoria in light and dark: a larger difference in light points towards a constrictor problem; a larger difference in dark points towards a dilator problem. This is a clue, not a complete diagnosis.

Figure reference

B: pupil reflex, anisocoria and Adie references. Original response matrix.